MCP by example

Conceiving, building, and testing MANNA

NASA AI/ML STIG seminar
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Talk Overview

- What is MCP?
- How to build an MCP server (demo)
- MCP in astronomy (MANNA)
- Assessing MCP quality

AI & Software Jargon

- **AI Agent**: Model (LLM) + Harness (claude code, cursor, etc)
- **Protocol**: standardized set of rules that governs how data is transmitted, received, and interpreted between devices on a network
- **Program**: a sequence of instructions written in a programming language that directs a computer to perform a specific task
- **Function**: a callable unit of code that does a specific task
- **API** (Application Programing Interface): mechanisms that enable two software components to communicate with each other using a set of definitions and protocols
- **Server**: A server is a role, that describes either a software program or the physical computer it runs on

MCP

What is it?

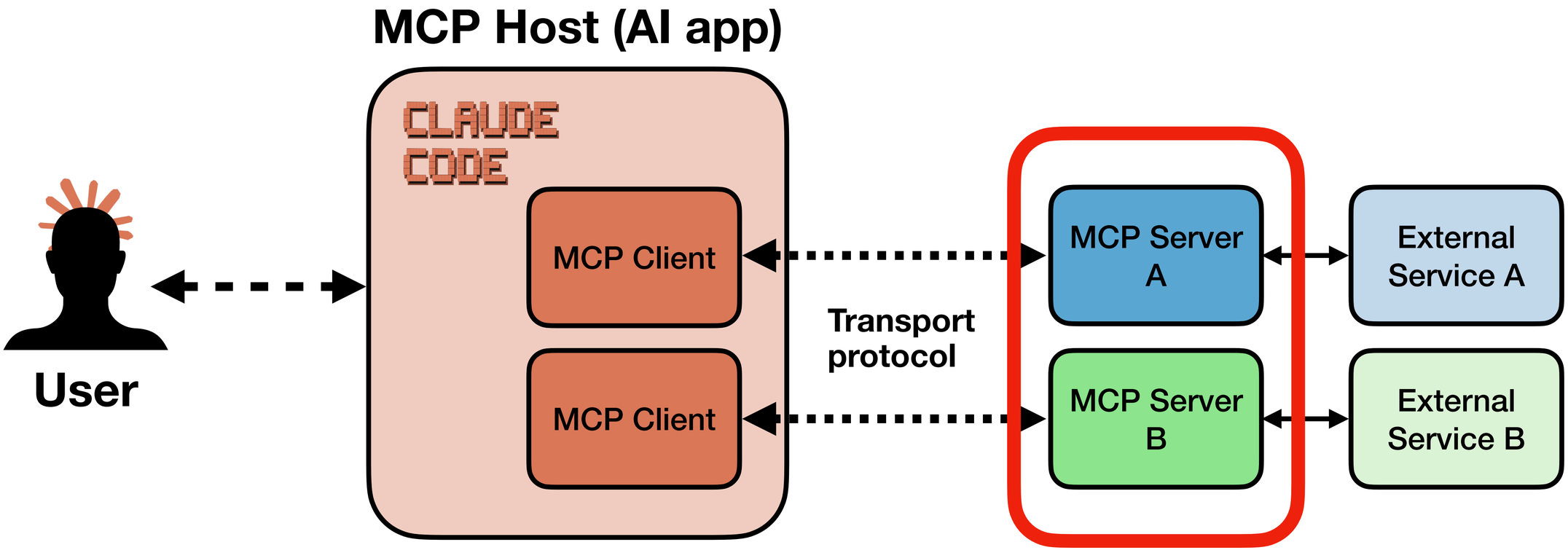
- **Model**: designed for LLMs and AI agents
- **Context**: supplies relevant information, beyond training data
- **Protocol**: standardized set of rules that govern how devices communicate and exchange data over a network
- Putting it all together… *MCP is an open-source standard [designed to connect] AI applications to external systems*.

Anthropic (2024); modelcontextprotocol.io

MCP: Components

What does it consist of?

- MCP Components: **hosts**, **clients**, and **servers**, and a **transport protocol**



MCP Servers

What is an MCP server?

- MCP servers are… servers!
- Consist of three features: **tools**, **resources**, and **prompts**
- Tools are functions, commonly used to build and send API requests
- Stateless
- Local and remote deployment options

MCP Specification (rev. 2026-07-28)

MCP Servers

What do we use them for?

- *“Common examples include file system servers for document access, database servers for data queries, GitHub servers for code management…”*
  - modelcontextprotocol.io
- Why not just have an AI agent access API directly?
- MCP servers frequently wrap APIs into a standardized layer for LLMs
- In practice, MCP servers are particularly useful for web services that…
  i. *require authentication,*
  ii. *have limited/outdated public online documentation,*
  iii. *or whose usage is contingent on significant contextual knowledge.*

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Astronomical Data Archives

- ☑ *authentication for proprietary/user specific data*
- ☑ *exhaustive documentation is challenging to maintain*
- ☑ *are generally used within complex workflows by astronomers*

MCP Servers

How do we build them?



- Typically written in TypeScript or Python
- Packages
  - **mcp:** The official Python SDK `❯ uv pip install mcp`
  - **fastmcp:** wrapper around **mcp** with more functionality



- Claude Skill: **mcp-server-dev** plugin
- MCP Inspector

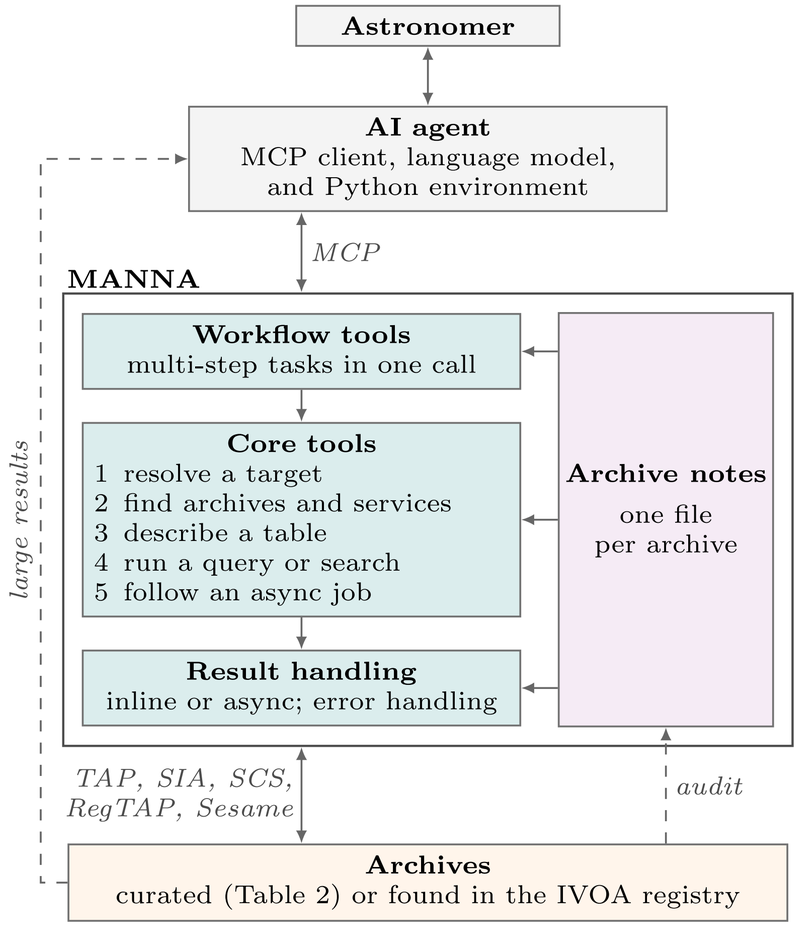
github.com/modelcontextprotocol/python-sdk · github.com/PrefectHQ/fastmcp · github.com/anthropics/claude-plugins-official · github.com/modelcontextprotocol/inspector

Demo

MANNA

MCP Architecture for NOIRLab, NRAO, and Additional Archives

- Flexible MCP server supporting AI agent access to IVOA-based archives
- Module based archive support: Astro Data Lab, ALMA, CADC, ESO, Gaia, and SDSS (NRAO paused)
- 15 tools mirroring steps in an astronomer's archive access workflow
  - Archive discovery, table inspection, querying, asynchronous job management



Gause et al. (2026, in prep)

MANNA: Evaluation

MCP Evaluation

- **Program I, Tool Use:** *does a model use MANNA's tools correctly?*
  - 14 tasks across four tiers, graded on tools called, arguments, and answer
- **Program II, Value Added:** *is a model better off with MANNA than without it?*
  - 23 tasks run with and without MANNA, comparing accuracy, cost, and time
  - Duplicate “capped “round with limits applied (max of 600s and 20 turns per task)
- **Program III, Note Audit:** *is what MANNA says about each archive still true?*
  - Scripted checks with no model, re-run against the live archives

Gause et al. (2026, in prep)

MANNA: Evaluation

Overview

- Programs I and II

- 4 models tested
  - Nemotron 3 Super (open-weight)
  - Haiku 4.5
  - Sonnet 5
  - Opus 4.8

- Claude Code harness





Gause et al. (2026, in prep)

MANNA: Evaluation

Program I

- Does a model use MANNA's tools correctly?

- **Tier 1, Tool Selection**
- **Tier 2, Task Completion**
- **Tier 3, Pitfalls**
- **Tier 4, Robustness**

Table 3. Program I: Tool-Use Evaluation.

Model	T1	T2	T3	T4	All
Nemotron 3 Super	4/5	3/4	2/2	3/3	12/14
Haiku 4.5	4/5	4/4	2/2	2/3	12/14
Sonnet 5	5/5	4/4	2/2	3/3	14/14
Opus 4.8	5/5	4/4	2/2	3/3	14/14

Gause et al. (2026, in prep)

MANNA: Evaluation

Program II

- Does MANNA improve agentic performance and accuracy?

Table 4. Program II: Value-Added Measurement.

	Tasks passed		Without vs. with MANNA, per solved task			
Model	With	Without	Calls	Tokens	Cost	Time
Nemotron 3 Super	21.0	20.3	15.5×	4.0×	···	8.1×
Haiku 4.5	21.7	15.0	3.1×	2.8×	6.7×	2.3×
Sonnet 5	22.0	21.3	1.1×	1.0×	1.3×	2.2×
Opus 4.8	23.0	23.0	2.3×	2.4×	2.4×	3.4×

Gause et al. (2026, in prep)

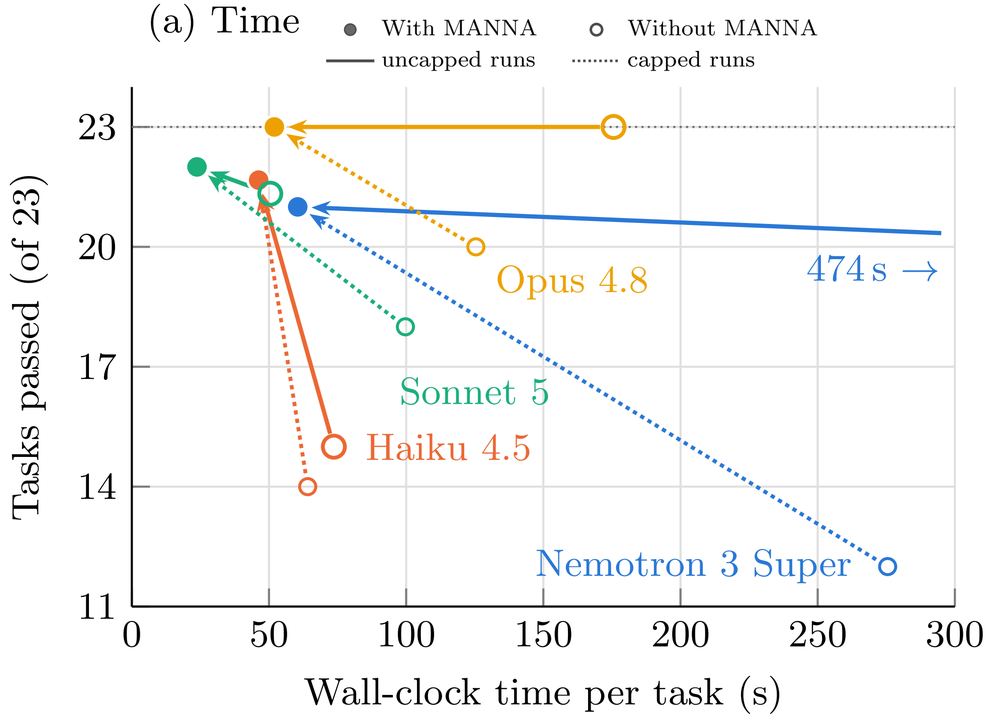
MANNA: Evaluation

Program II: Speed and Accuracy

- Does MANNA improve agentic performance and accuracy?

Table 4. Program II: Value-Added Measurement.

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- MANNA speeds up AI agent task completion for all models, and substantially improves accuracy for Haiku 4.5
- Substantial improvements in speed and accuracy for all models under limits

Gause et al. (2026, in prep)

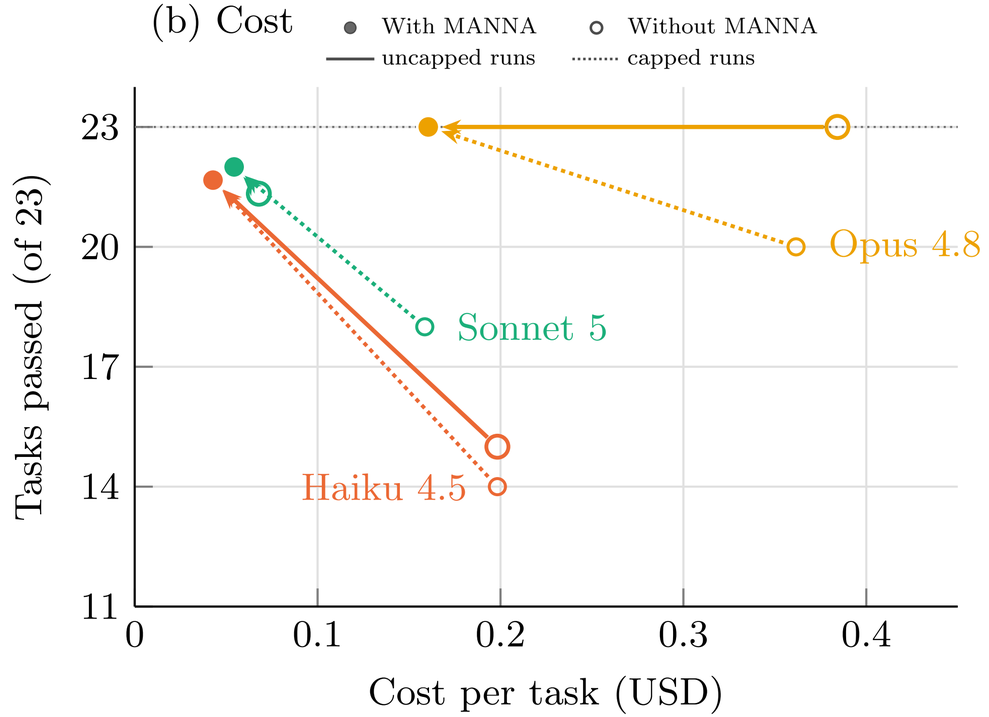
MANNA: Evaluation

Program II: Cost

- Does MANNA affect agents' cost?

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- MANNA decreases cost of agent task completion for all models (no open-weight)
- Applying limits magnified effect

Gause et al. (2026, in prep)

MANNA

Deployment and Installation

- Testing Astro Data Lab MANNA + JupyterAI deployment
- Docs at manna.readthedocs.io
- Available for local use
- Installable from
  - CosmicAI GitHub: source code or container (ghcr)
  - PyPI: **manna-mcp**
  - MCP Registry and Marketplaces

Gause et al. (2026, in prep)

Conclusions

- MANNA narrowed the gap between models of varying capabilities, improving cost and time performance for all models and improving accuracy for lower capability models
- Enables the reliable use of non-frontier models (open-weight, haiku)

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- This work was supported by the National Science Foundation under Cooperative Agreement 2421782 and the Simons Foundation award MPS-AI-00010515, through the CosmicAI seed funding project STABLE

Questions?

CosmicAI

SIMONS FOUNDATION

Gause et al. (2026, in prep)

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Additional Slides



MCP Servers

Best Practices

- Intent based tooling
- Prioritize docstrings and schemas
- Structured error handling and reporting
- Logging
- Security (especially for API requests)
- Choose the right transport (stdio vs. Streamable HTTP)

MANNA: Evaluation

Overview

- Nemotron 3 Super (open-weight)
  - Model ID: nvidia/NVIDIA-Nemotron-3-Super-120B-A12B-FP8
  - Architecture: open-weight mixture-of-experts model with 120B total parameters, 12B of them active.
  - Precision: NVIDIA's FP8-quantized release.
  - Self-hosted with vLLM
- Haiku 4.5 (no effort)
- Sonnet 5 (effort default: high)
- Opus 4.8 (effort default: high)

- Harness: Claude Code v2.1.259 against MANNA v0.10.0

Gause et al. (2026, in prep)

MANNA: Evaluation

Program I: Question Examples

- Does a model use MANNA's tools correctly?
- **Tier 1, tool selection**: picks the right tool with the right arguments for a single request.
  - "Are there any images at RA=202.47, Dec=47.19 in the CADC archive?"
- **Tier 2, task completion:** chains tools together to finish a real workflow.
  - "I need stellar proper motions from Gaia DR3. Discover the right TAP service, inspect it, then run a small query returning source_id, pmra, pmdec."
- **Tier 3, pitfalls:** avoids known archive-specific mistakes.
  - "How many distinct ALMA observations are there of the galaxy NGC 253? Be careful about how ALMA counts rows."
- **Tier 4, robustness:** recovers from failures instead of giving up or making things up.
  - "Query the 'FooBar Deep Survey' archive for galaxies brighter than 18th magnitude."

MANNA: Evaluation

Program II: Question Examples

- **1. Name resolution:** "What are the ICRS coordinates (decimal degrees) of the Whirlpool Galaxy (M51)?"
- **2. Catalog query:** "Find Gaia DR3 sources within 0.02 degrees of the galaxy M87 and report how many there are, plus the proper motion (pmra, pmdec) of one or two example sources."
- **3. Archive-specific pitfall (Data Lab geometry):** "Using the NOIRLab Astro Data Lab NSC DR2 catalogue (nsc_dr2.object), how many sources lie within 0.02 degrees of RA=60.0, Dec=-30.0?" To pass, the model has to use q3c_radial_query, because ADQL CONTAINS fails on Data Lab.
- **4. Archive-specific pitfall (ALMA counting):** "How many DISTINCT ALMA observations (downloadable Member OUS datasets) are within 0.1 degrees of the galaxy M83? ALMA rows are finer-grained than observations, so count distinct datasets, not raw rows."
- **5. Cross-archive survey:** "Give a cross-archive summary of what data is available within 0.1 degrees of the radio galaxy Centaurus A (NGC 5128): for each archive you can reach, report whether it has data there and roughly how many observations or sources."

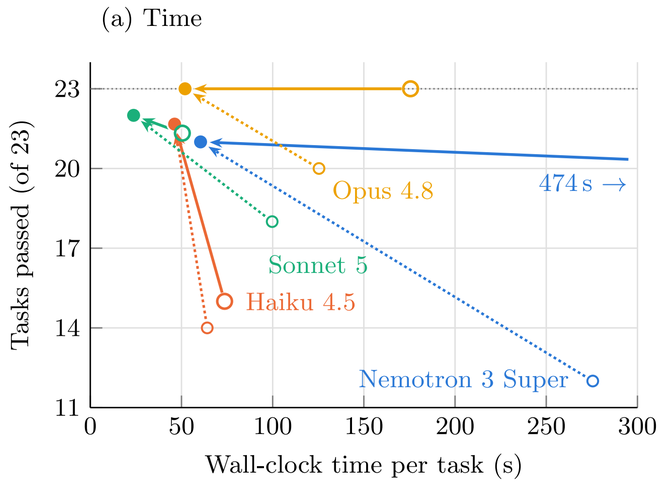
MANNA

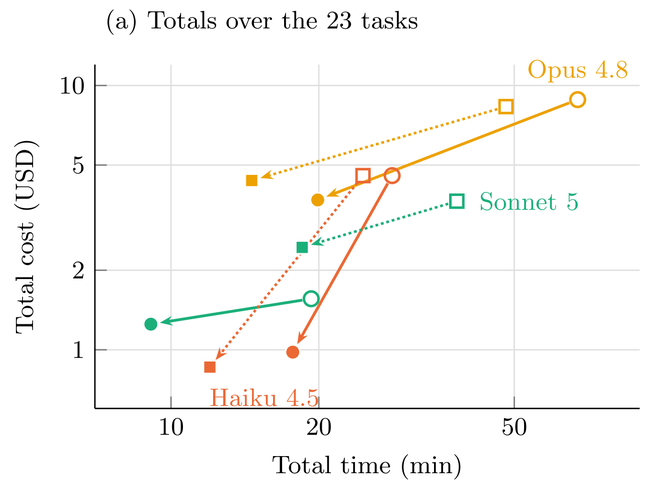
MCP Evaluation: Program II

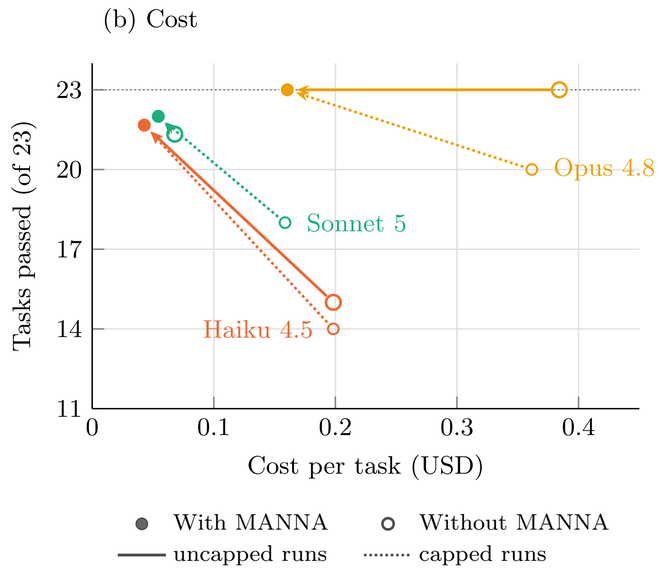
- Does MANNA improve model performance and accuracy?

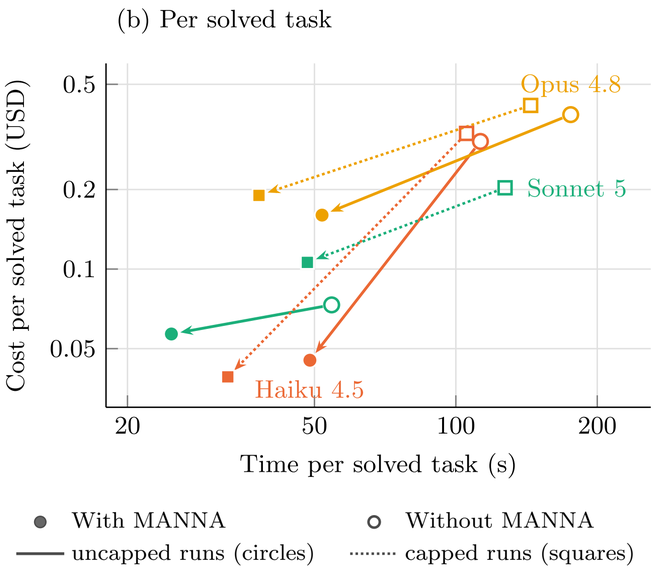
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Sonnet 5	22.0	21.3	1.1×	1.0×	1.3×	2.2×
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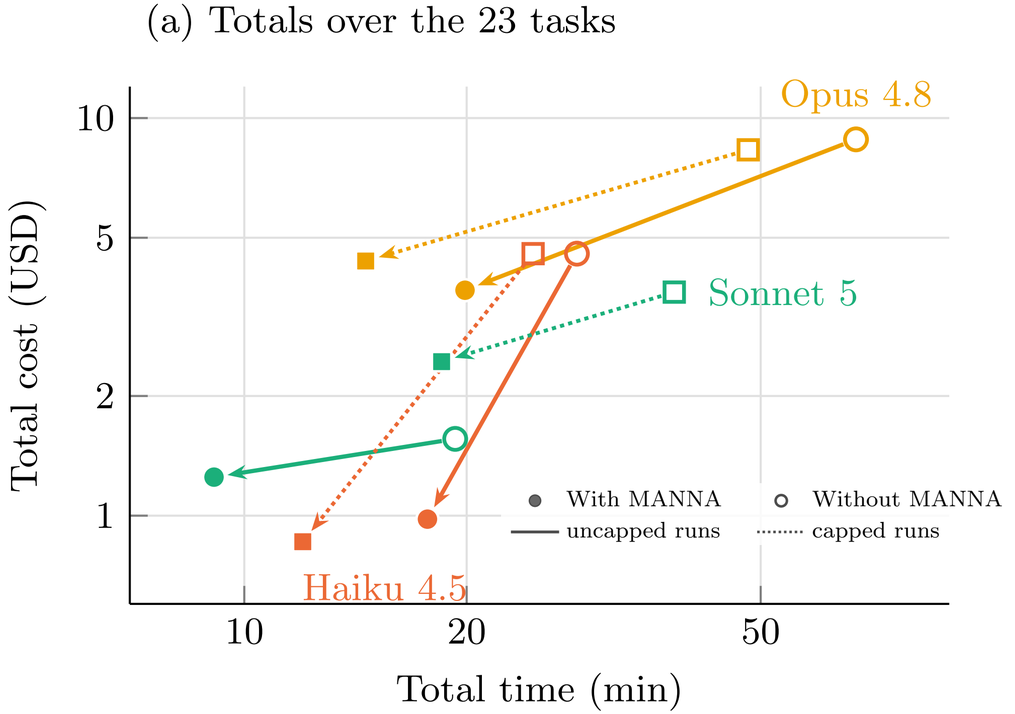
MANNA: Evaluation

Program II: Time and Cost Totals

- Does MANNA affect agents' cost?

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- Decreased total time by at least a factor of 2
- Decreased total cost by a factor of 1.3 - 6.7

Gause et al. (2026, in prep)

A. PROGRAM II: FULL RESULTS

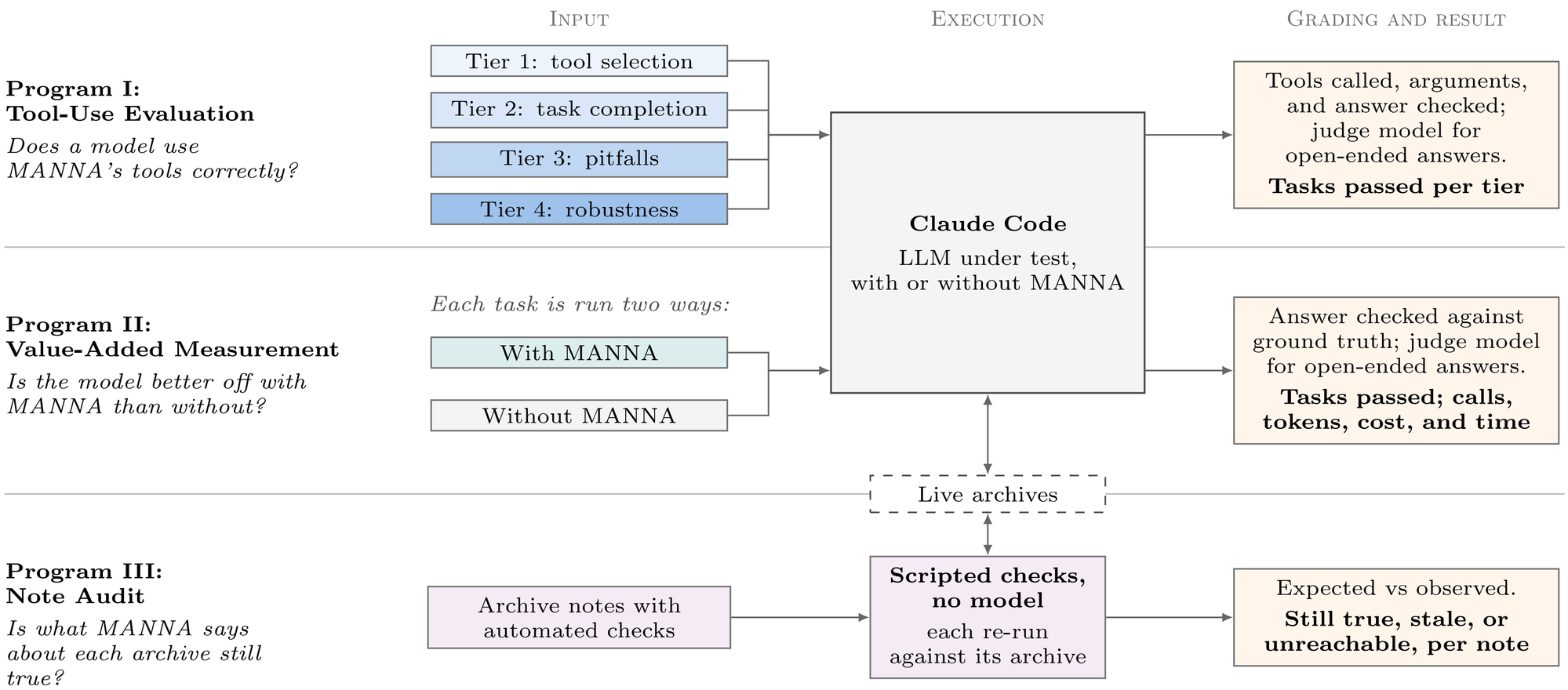
Table 5. Program II: Value-Added Measurement, full results for every model, harness, approach, and budget on the 23 Program II tasks.

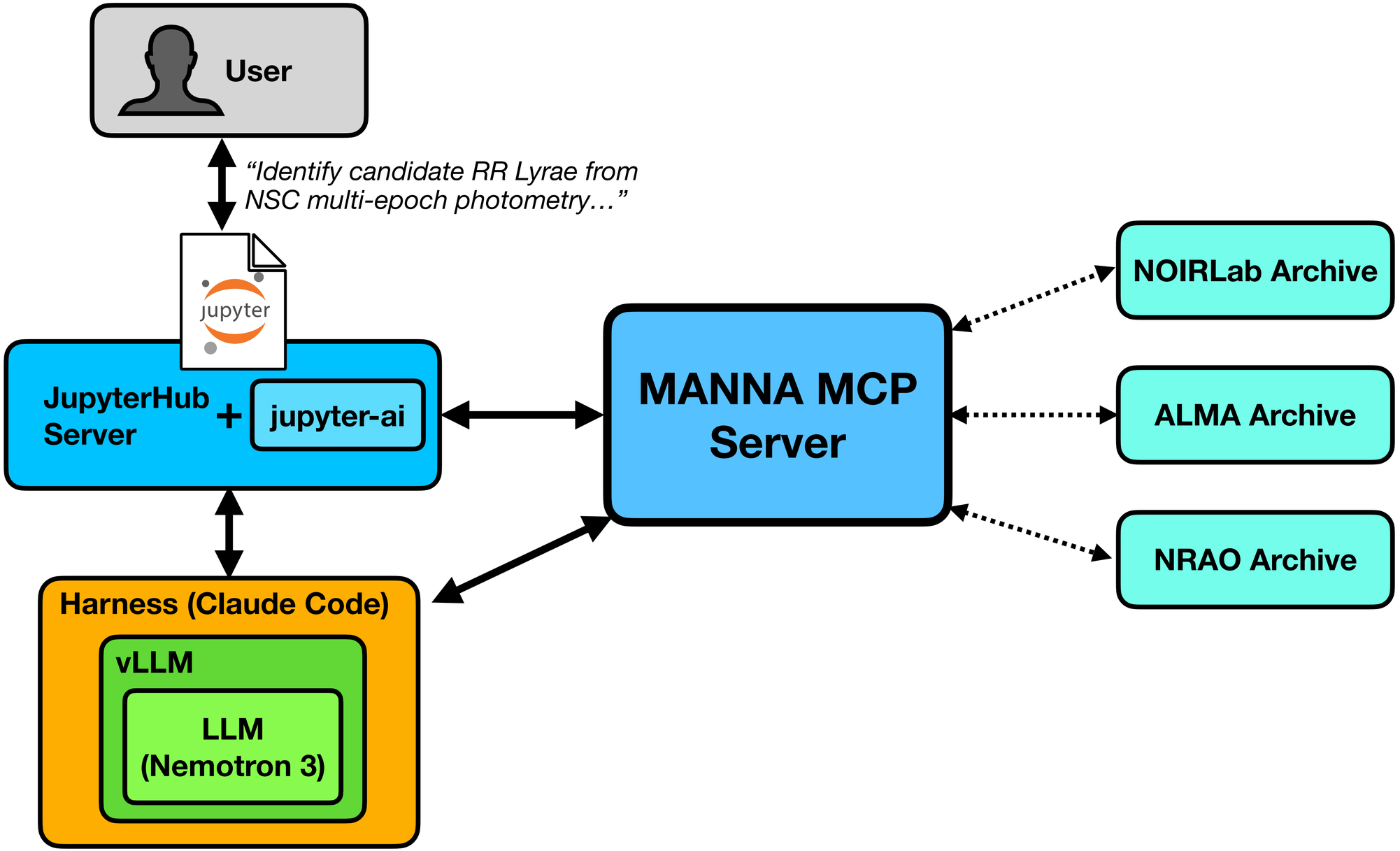
Model	Approach	Pass	Give-ups	Tool calls	Errors	Tokens (M)	Cost	Time (min)	Per task (s)
				Claude Code					
Nemotron 3 Super	With, uncapped	21.0 (20–22)	0	73	1	3.40	···	23.2	60
	Without, uncapped	20.3 (19–22)	0.3	1098 (15.0×)	175	13.18 (3.9×)	···	181.7 (7.8×)	474
	With, capped	22	0	62	0	2.93	···	19.6	51
	Without, capped	12	11	143 (2.0×)	37	3.36 (1.0×)	···	105.6 (4.6×)	276
Haiku 4.5	With, uncapped	21.7 (21–22)	0	100	1	3.15	$0.98	17.7	46
	Without, uncapped	15.0 (14–17)	0	215 (2.1×)	8	6.05 (1.9×)	$4.56 (4.6×)	28.2 (1.6×)	74
	With, capped	22	0	83	1	2.44	$0.86	12.0	31
	Without, capped	14	4	203 (2.0×)	6	5.51 (1.7×)	$4.56 (4.6×)	24.6 (1.4×)	64
Sonnet 5	With, uncapped	22.0 (21–23)	0	80	0	1.79	$1.25	9.1	24
	Without, uncapped	21.3 (21–22)	0	85 (1.1×)	1	1.67 (0.9×)	$1.56 (1.2×)	19.3 (2.1×)	50
	With, capped	23	0	91	0	3.63	$2.44	18.5	48
	Without, capped	18	4	149 (1.9×)	7	5.51 (3.1×)	$3.65 (2.9×)	38.2 (4.2×)	100
Opus 4.8	With, uncapped	23.0 (23–23)	0	80	0	2.00	$3.69	19.9	52
	Without, uncapped	23.0 (23–23)	0	181 (2.3×)	14	4.83 (2.4×)	$8.84 (2.4×)	67.3 (3.4×)	176
	With, capped	23	0	93	0	2.41	$4.37	14.6	38
	Without, capped	20	3	192 (2.4×)	12	3.13 (1.6×)	$8.32 (2.3×)	48.1 (2.4×)	125
				Custom agent loop					
Nemotron 3 Super	With MANNA	21	0	75	8	1.23	···	16.9	44
	Bare TAP client	8	13	316 (4.2×)	178	1.92 (1.6×)	···	69.0 (4.1×)	180
	Web only	14	7	229 (3.1×)	89	5.23 (4.2×)	···	55.8 (3.3×)	146
Haiku 4.5	With MANNA	23	0	60	1	1.01	···	8.9	23
	Bare TAP client	16	3	274 (4.6×)	203	1.07 (1.1×)	···	20.7 (2.3×)	54
	Web only	19	1	198 (3.3×)	74	2.43 (2.4×)	···	21.8 (2.4×)	57
Sonnet 5	With MANNA	23	0	60	0	1.51	···	13.2	34
	Bare TAP client	22	0	167 (2.8×)	59	0.96 (0.6×)	···	64.3 (4.9×)	168
	Web only	20	2	212 (3.5×)	81	3.72 (2.5×)	···	59.0 (4.5×)	154
Opus 4.8	With MANNA	22	···	83	···	···	···	···	···
	Bare TAP client	19/19	···	55	···	···	···	···	···

NOTE—Runs, logs, and analysis scripts: doi:10.5281/zenodo.23017197 (D. Gause 2026). Pass is out of 23 (range across runs in parentheses); other parentheses give the ratio to the model's first row in the same harness. Give-ups are tasks that hit a limit, scored as fails. Totals are over all 23 tasks; Per task is mean wall-clock time. Tokens are input tokens, including cached ones and Claude Code's system prompt, and are comparable only within a harness. *Claude Code*: uncapped rows are the mean of three runs, each allowed 1 h and 200 turns per task (give-ups and errors are means); capped rows are one run limited to 600 s and 20 turns per task, what a client with default limits sees. Without MANNA, the agent keeps its built-in tools (Bash, WebFetch, WebSearch); cost is Claude Code's estimate at list prices (§3.2). *Custom agent loop* (one run): 20 steps, no time limit, no cost recorded; Opus used adaptive thinking; the MANNA approach adds an `execute_python` tool for fetch recipes; the bare TAP client is PyVO with no archive notes. The Opus run stopped after 19 bare-TAP tasks when connections to CADC and ALMA hung, before the web-only approach ran; its numbers come from the run log, which records neither errors, tokens nor time on a comparable basis.

MANNA

MCP Evaluation




User
"Identify candidate RR Lyrae from NSC multi-epoch photometry…"
jupyter
JupyterHub Server
+
jupyter-ai
MANNA MCP Server
NOIRLab Archive
ALMA Archive
NRAO Archive
Harness (Claude Code)
vLLM
LLM (Nemotron 3)

“What DECam data exist for NGC 253?”

Agentic Reasoning Chain

MANNA Tool Calls

1 *“Optical — that’s NOIRLab’s own holdings. Which archive, and what should I know first?”*

waveband=“optical" → datalab

vo_archive_list

2 *“Any spatial query needs coordinates, not a name.”*

NGC 253 → 11.888°, –25.288°

vo_target_resolve

3 *“Data Lab doesn’t translate ADQL geometry. Check the table before I write a query that silently fails.”*

archive="datalab", table="nsc_dr2.object"

vo_schema_describe

4 *“So: Q3C, not CONTAINS — and it needs the = 't'.”*

6,150 objects

vo_tap_query

5 *“Catalogue done. Is there imagery too?”*

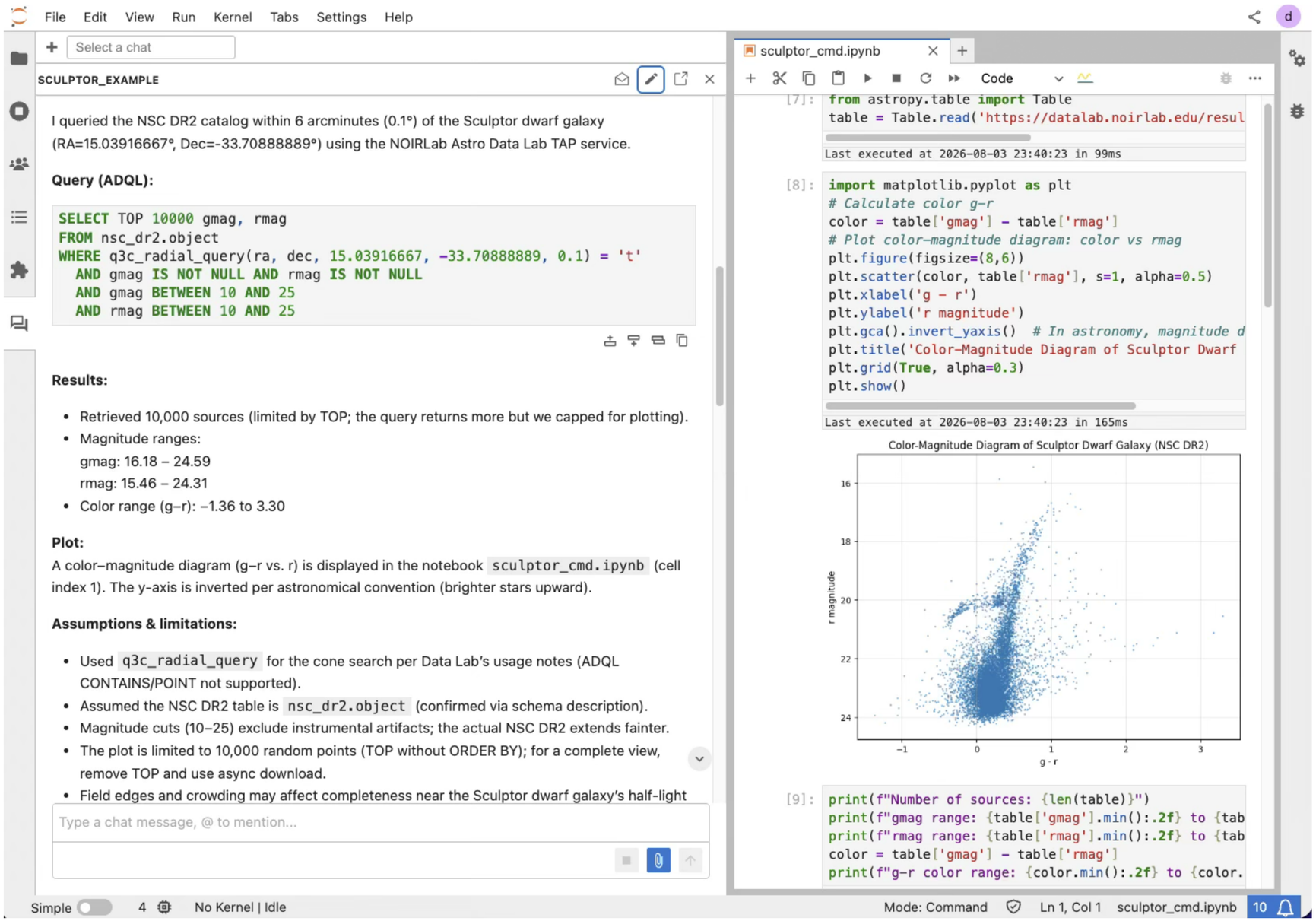
31 DECam cutouts

vo_sia_search

ANSWER: 6,150 NSC DR2 sources within 3′, 31 DECam science images in grizY, from DES DR1; NGC 253 is well covered optically, in i and z especially.

Astro Data Lab jupyter-ai + MANNA Example

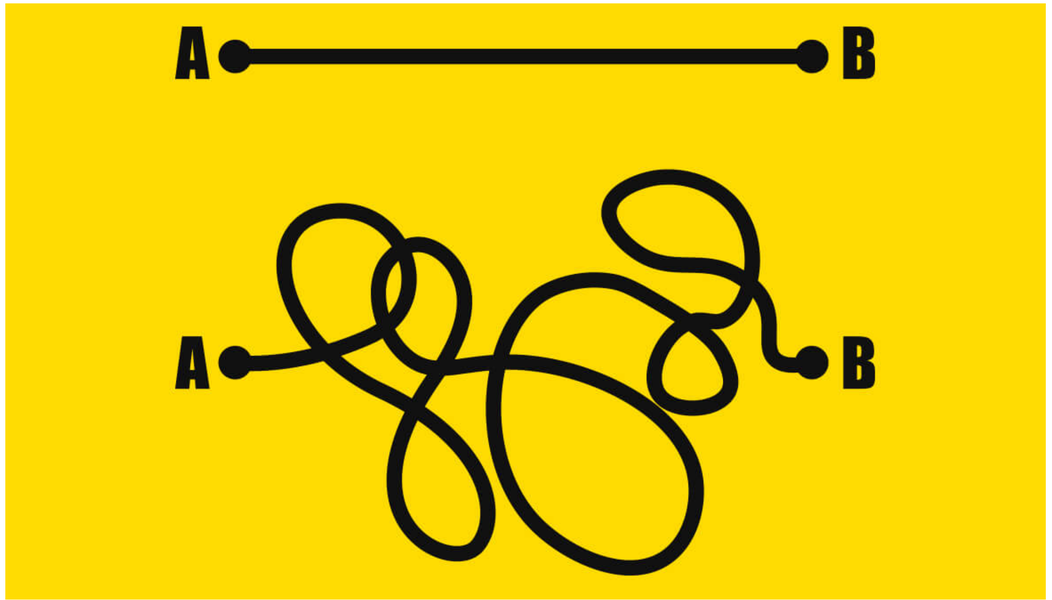
“Using Astro Data Lab, get NSC DR2 sources within 6 arcminutes of the Sculptor dwarf galaxy and plot a color–magnitude diagram.”



MCP: Why?

My frontier model can figure it out; why do we still need MCP?

- Most direct route to a goal, don’t rely on iterative plug&play. In short, $$
- Establish the *best* way to accomplish a task instead of letting an agent find *some* way to accomplish it



MCP: Why?

Why would I use MCP instead of a skills file?

- MCP is not always the right answer!
  - Do you have control over the client side?
  - What is the level of complexity of your need?
- MCP tools can be *dynamic*, skills files will always be *static*
- Determinism: it's actual code! Tool code runs the same every time, skills are interpreted
- Server benefits: error handling, input/output validation
- Server-side compute; can offload heavy compute tasks from AI agent to dedicated server